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Efficient anomaly detection in tabular cybersecurity data using large language models

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In cybersecurity, anomaly detection in tabular data is essential for ensuring information security. While traditional machine learning and deep learning methods have shown some success, they continue to face significant challenges in terms of generalization. To address these limitations, this paper presents an innovative method for tabular data anomaly detection based on large language models, called "Tabular Anomaly Detection via Guided Prompts" (TAD-GP). This approach utilizes a 7-billion-parameter open-source model and incorporates strategies such as data sample introduction, anomaly type recognition, chain-of-thought reasoning, multi-turn dialogue, and key information reinforcement. Experimental results indicate that the TAD-GP framework improves F1 scores by 79.31%, 97.96%, and 59.09% on the CICIDS2017, KDD Cup 1999, and UNSW-NB15 datasets, respectively. Furthermore, the smaller-scale TAD-GP model outperforms larger models across multiple datasets, demonstrating its practical potential in environments with constrained computational resources and requirements for private deployment. This method addresses a critical gap in research on anomaly detection in cybersecurity, specifically using small-scale open-source models.

Keywords Anomaly detection, Large language models, Network security, Prompt engineering, Tabular data

In cybersecurity, tabular data is commonly used to capture critical information such as network traffic, user behavior, and system logs. Typically stored in matrix format, with each row representing a sample and each column corresponding to a feature, this data is crucial for ensuring network security. However, as cyberattacks become increasingly sophisticated, traditional rule-based detection methods have become inadequate. Consequently, researchers have turned to advanced anomaly detection techniques like machine learning and deep learning, which, while successful to some extent, still face challenges related to generalization[1].

Traditional machine learning and deep learning models often struggle with unseen data distributions and are prone to overfitting, limiting their ability to detect novel or sophisticated attacks. This lack of generalization presents a major challenge for network anomaly detection, necessitating the development of new approaches.

Large language models (LLMs), on the other hand, have shown stronger generalization capabilities. They have achieved notable success in natural language processing tasks and show potential in cybersecurity applications. By leveraging techniques such as zero-shot, one-shot, and few-shot learning, LLMs can address complex structures and relationships, making them a promising solution to the generalization problem in anomaly detection[2]. However, LLMs are typically large, requiring significant computational resources, which limits their application in resource-constrained or private deployment scenarios.

Most current research focuses on large-scale models like GPT-3.5-turbo. Although these models are powerful, their high computational and memory demands hinder practical application. To address this, the present study proposes an innovative approach by using a 7-billion-parameter open-source model within a guided prompt framework called "Tabular Anomaly Detection via Guided Prompts" (TAD-GP). This approach aims to achieve efficient anomaly detection in resource-constrained environments. Notably, experimental results indicate that TAD-GP outperforms larger models such as GPT-3.5-turbo in certain anomaly detection tasks. The study addresses a research gap by demonstrating the effectiveness of small-scale open-source models for cybersecurity anomaly detection.

This study's objective is to design a guided prompt framework tailored for small models, assess its performance in anomaly detection tasks, and explore the broader applications of LLMs in cybersecurity. The significance of this research lies in its ability to provide a novel and efficient anomaly detection method that can uncover hidden patterns and behaviors in tabular data, offering new insights for network security. Optimizing the prompt framework is expected to improve detection accuracy, reduce false positives and negatives, and

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enhance the overall practicality of the model. Furthermore, using small-scale open-source models supports private deployment, enhancing energy efficiency, deployment flexibility, and privacy protection.

To aid readers, the structure of this paper is outlined as follows:

Introduction Discusses the importance of anomaly detection in tabular data for cybersecurity, identifies the limitations of traditional methods, and introduces the TAD-GP framework as a solution.

Related work Reviews traditional anomaly detection methods and examines the potential of LLMs in cybersecurity, highlighting challenges and research progress, particularly in applying small-scale open-source models in resource-limited environments.

TAD-GP framework Provides a detailed description of the TAD-GP framework, including key strategies such as data sample introduction, anomaly type recognition, chain-of-thought reasoning, multi-turn dialogue, and information reinforcement.

Experimental design Outlines the experimental objectives, dataset selection, models, evaluation metrics, and setup, validating the TAD-GP framework through a series of experiments.

Experimental results and discussion Analyzes experimental findings, examines the impact of various strategies on model performance, and discusses the framework's potential applications, challenges, and future research directions.

Conclusion Summarizes the innovation and practicality of the TAD-GP framework, emphasizing its advantages in resource-constrained environments, including energy efficiency, flexible deployment, and privacy protection support.

Related work

Anomaly detection in tabular data is crucial for network security, as it helps identify abnormal behaviors, unusual patterns, and potential threats within structured datasets.

Traditional rule-based detection methods rely on predefined patterns or thresholds to identify anomalies by comparing current data with established norms. However, as network attack strategies become increasingly sophisticated, the effectiveness of these methods has declined. In response, machine learning and deep learning approaches have gained prominence. These data-driven techniques analyze large volumes of historical data to detect abnormal patterns and behaviors, enabling the prediction and recognition of future anomalies.

Representative machine learning methods include:

One-Class Support Vector Machine (OC-SVM)[3]: OC-SVM models normal data and flags deviations from this learned model as anomalies; Isolation Forest[4]: This method assumes that anomalies exist in low-density regions of feature space and isolates them by constructing decision trees using an ensemble approach; Gaussian Mixture Model (GMM)[5]: GMM estimates the data's underlying distribution and labels points that deviate from this distribution as anomalies.

Deep learning-based approaches include:

Autoencoders[6]: These models compress normal data representations and detect anomalies based on high reconstruction errors; Generative Adversarial Networks (GANs)[7]: GANs model the distribution of normal data, identifying anomalies as samples that do not conform to this distribution; Attention-based Models[8]: Recent research has explored attention mechanisms to enhance interpretability and performance in anomaly detection for tabular data.

Recent developments in anomaly detection methods include:

Using labeled data to explain detected anomalies[9]; Proposing single-class adversarial learning for anomaly detection in tabular data[10]; Developing benchmarks like Adbench to evaluate anomaly detection methods comprehensively[11].

LLMs offer new possibilities for anomaly detection due to their exceptional generalization abilities. For instance, GPT-3.5-turbo, developed by OpenAI, is based on the Transformer architecture and optimized with hundreds of billions of parameters. Pre-trained on extensive datasets, GPT-3.5-turbo has demonstrated proficiency in various natural language processing tasks. By leveraging their deep language understanding, LLMs present innovative approaches to anomaly detection. Researchers are exploring how LLMs can analyze data more comprehensively to uncover hidden anomalous behaviors and improve detection accuracy and efficiency. Examples include:

Text-based anomaly detection: LLMs can be trained on large text datasets, such as financial reports, medical records, or network security logs[12], to learn data patterns and detect anomalies; Contextual anomaly detection: LLMs' ability to capture and understand context[13] helps differentiate between normal and abnormal patterns, reducing false positives; Active anomaly detection: LLMs can generate synthetic data or scenarios[14] to stress-test anomaly detection systems, aiding organizations in preparing for future anomalies or threats.

Tabular anomaly detection via guided prompting (TAD-GP)

This paper introduces a novel approach for anomaly detection in tabular network data, called the Tabular Anomaly Detection via Guided Prompts (TAD-GP) framework (Figs. 1 and 2). The framework is designed to improve the performance of small-scale large language models in anomaly detection tasks. TAD-GP incorporates several key techniques, including data sample introduction, fine-grained anomaly classification, chain-of-thought reasoning, multi-turn interaction, and key information reinforcement. These strategies collectively guide the model through a structured analytical process, resulting in more precise and efficient anomaly detection.

Efficient introduction of data samples

To improve the model's understanding of tabular data, the TAD-GP framework utilizes a strategy of selecting one normal and one anomalous sample at random. This helps the model quickly familiarize itself with key data

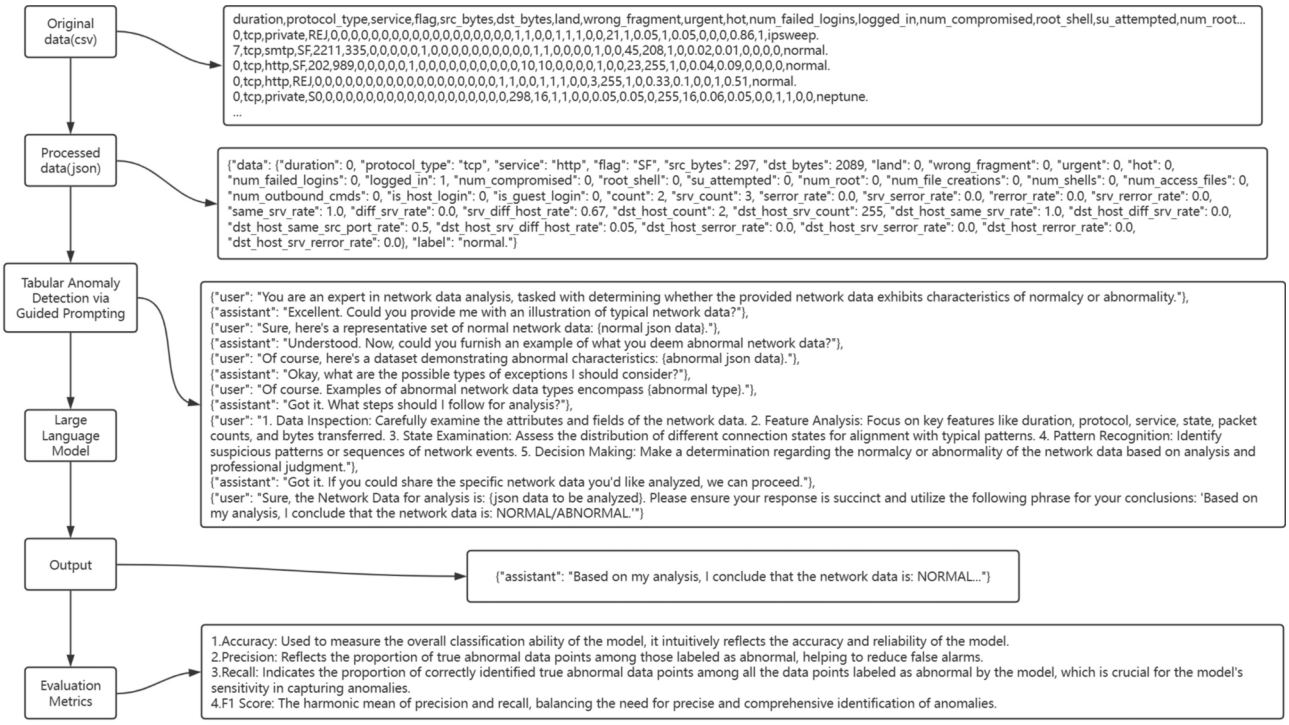

Fig. 1. Structured Processing Flowchart for Tabular Data Anomaly Detection. This figure outlines the complete process from raw data input to the final anomaly detection output. The data is first converted into JSON format and integrated into a specially designed multi-turn dialogue. The large language model then generates the corresponding linguistic analysis and anomaly detection results, which are followed by a performance evaluation of the model.

features and patterns (as shown in the green section of Fig. 2). The selected samples are converted into JSON format, facilitating the model's processing and understanding of the data's structure and attributes (as seen in the "Processed Data" section of Fig. 1). JSON, commonly used as input for large language models, provides structured information, simplifying the handling of complex feature relationships. This approach enhances the model's ability to recognize critical patterns early on, leading to faster and more accurate anomaly detection.

Fine-grained introduction of anomaly categories

The TAD-GP framework establishes a clear sequence of anomaly categories for each dataset, which guides the model in identifying various types of anomalous behavior. This approach utilizes predefined categories-such as denial-of-service (DOS), unauthorized access from a remote machine to a local machine (R2L), unauthorized access to local superuser privileges by a local unpivileged user (U2R), and surveillance and probing (PROBING) from the KDD Cup 1999 dataset-to facilitate more precise classifications during anomaly detection tasks (as illustrated in the blue section of Fig. 2). The introduction of fine-grained anomaly categories not only enhances the model's ability to recognize anomalies but also provides a more detailed classification framework for these tasks. This strategy reduces the risk of confusing different types of anomalies, thereby improving both detection accuracy and robustness.

"Part-first, whole-later" chain-of-thought strategy

To improve the interpretability of anomaly detection, the TAD-GP framework employs a "part-first, whole-later" chain-of-thought strategy. In this approach, the model begins by independently analyzing each feature to evaluate the likelihood of deviation from normal ranges. It then synthesizes the interactions among multiple features to identify potential anomaly patterns, as illustrated in the purple section of Fig. 2.

For instance, in the Data Inspection section, we guide the model to focus on individual features of the data instances. In the Pattern Recognition section, we direct the model to consider interactions among multiple features. Finally, in the Decision Making section, we assist the model in making the final judgment.

This strategy enhances understanding of the complex relationships among data features, enabling the model to detect not only explicit anomalies but also more subtle combinations of anomaly patterns. This significantly improves its detection capability. Particularly in high-dimensional and complex data scenarios, the strategy allows the model to analyze feature dependencies in greater detail, leading to more comprehensive and accurate judgments.

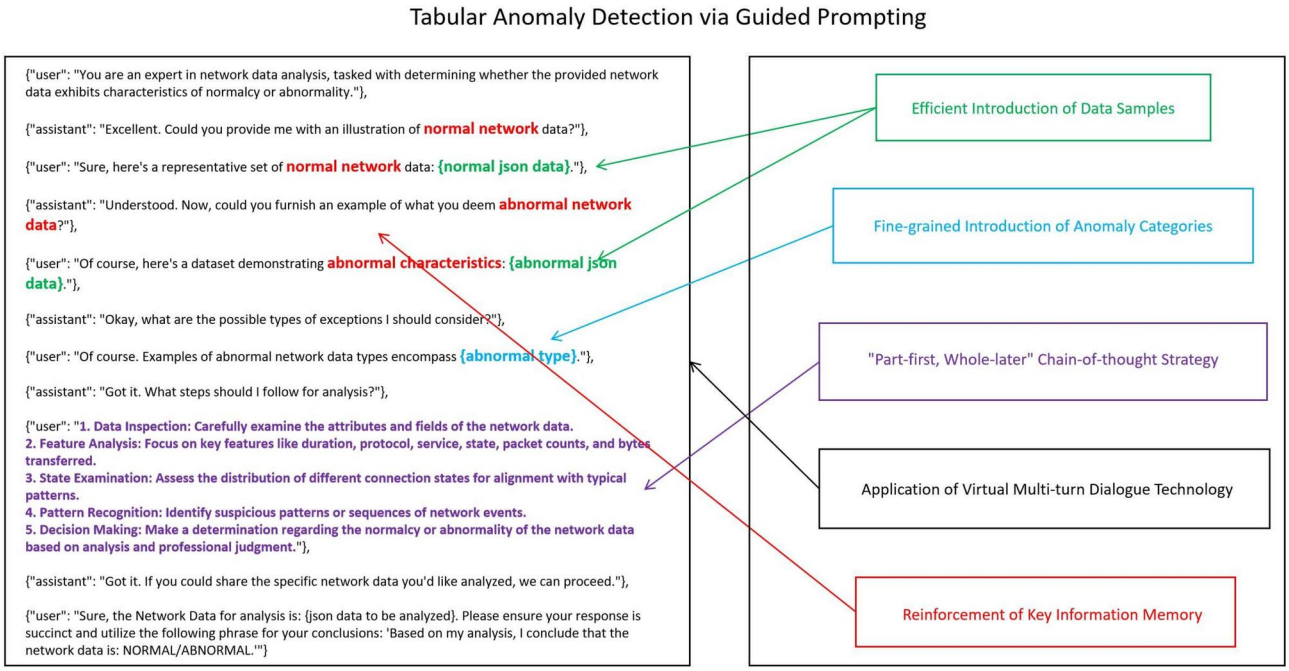

Fig. 2. Architecture of the Tabular Anomaly Detection via Guided Prompting. This figure presents the technical application of the TAD-GP framework for tabular anomaly detection, detailing key components such as data sample introduction, fine-grained anomaly type identification, chain-of-thought processing, multi-turn dialogue, and key information reinforcement. In practice, the green placeholders {normal json data}, {abnormal json data}, and the black {json data to be analyzed} should be replaced with actual data, following the format of the processed data in Fig. 1. The blue placeholder {abnormal type} should be substituted with the relevant anomaly types from the dataset. For instance, in the KDD Cup 1999 dataset, these types include denial-of-service (DOS), unauthorized access from a remote machine to a local machine (R2L), unauthorized access to local superuser privileges by a local unpivileged user (U2R), and surveillance and probing (PROBING).

Application of virtual multi-turn dialogue technology

To simulate the thinking process of experts engaged in complex reasoning tasks, the TAD-GP framework incorporates a virtual multi-turn dialogue mechanism. Each round of dialogue consists of a user posing a question and the corresponding assistant providing an answer ({"user": "Sure, here's a representative set of normal network data: {normal json data}."}, {"assistant": "Understood. Now, could you furnish an example of what you deem abnormal network data?"}). Within each dialogue round, the model focuses on a specific aspect of the anomaly detection task, moving from data examples to feature analysis while gradually deepening its understanding of anomaly patterns (refer to the black section in Fig. 2).

This multi-turn dialogue alleviates the cognitive burden on the model during single-instance analysis. Additionally, it fosters more precise reasoning and judgment regarding anomalies through a structured guidance approach. The introduction of virtual dialogue, in contrast to real dialogue, enhances the model's reasoning speed. Through this artificially constructed interaction process-since the assistant's responses are designed by humans and termed "virtual"-the model is effectively guided to engage in deep thinking, ensuring that its analytical process aligns with predefined expectations.

Reinforcement of key information memory

The TAD-GP framework introduces a key information memory reinforcement mechanism that ensures the model consistently retains critical data points throughout the multi-turn dialogue process (see the red section in Fig. 2). For example, in the first round of dialogue, the model is prompted to obtain a normal data sample with the interaction {"assistant": "Excellent. Could you provide me with an illustration of normal network data?"}. In the second round, the normality of this data is further emphasized with the response {"user": "Sure, here's a representative set of normal network data: {normal json data}."}.

This mechanism reinforces key information by consistently emphasizing important details across various dialogue rounds. This approach prevents the model from overlooking significant elements and ensures consistency throughout the anomaly detection process. By repeatedly highlighting essential features, the model can strengthen its memory of critical anomaly patterns, thereby improving overall detection accuracy. This memory reinforcement mechanism is particularly beneficial in scenarios involving complex feature interrelations, as it helps the model retain key information and enhances its reliability in challenging data environments.

Experimental design

This study aims to evaluate the potential of large-scale language models for tabular data anomaly detection and to validate their effectiveness in smaller models through the introduction of an innovative TAD-GP framework. To thoroughly assess the model's performance under various conditions, the experimental design includes research across multiple aspects.

Model and dataset selection

In this experiment, the Gemma-7b-it model was chosen as the base model, and three widely used benchmark datasets in cybersecurity were selected: KDD Cup 1999, CICIDS2017[15], and UNSW-NB15[16–20]. Each dataset encompasses normal traffic alongside various types of attack behaviors, rendering them suitable for evaluating network intrusion detection and anomaly detection tasks.

Data preprocessing

Data cleaning Remove duplicates and address missing values in the dataset to maintain data integrity and consistency.

Data balancing Utilize sampling techniques to achieve a balanced ratio of normal to anomalous data, thereby preventing class imbalance from adversely affecting the model's training process.

Data format conversion Convert tabular data into a JSON format that is compatible with large language models, facilitating efficient parsing of structured data by the model.

Evaluation metrics

This study employs the following commonly used classification metrics to comprehensively assess the model's performance under various experimental conditions:

Accuracy This metric measures the overall correctness of the model's classifications across all data samples.

Precision This metric evaluates the proportion of actual anomalies among the samples that the model predicts as anomalous.

Recall This metric assesses the model's ability to detect actual anomalous data.

F1 score This metric represents the harmonic mean of precision and recall, providing insight into the balance between these performance metrics.

Experiment groups

Experiment 1: impact of chain-of-thought strategy

This experiment examines the effectiveness of the chain-of-thought strategy in anomaly detection tasks. A "part-first, whole-later" guided approach is used, where the model is incrementally provided with partial information during the inference process. This method encourages Chain-of-thought reasoning on complex data features. The experiment compares the model's performance with and without the chain-of-thought strategy to determine if it enhances the model's reasoning ability in anomaly detection.

Experiment 2: impact of multi-turn dialogue strategy

This experiment tests the effectiveness of multi-turn dialogue in improving the model's understanding of complex data features. A multi-turn dialogue scenario is designed, where key data features are progressively introduced through multiple rounds of interaction. The experiment compares the model's performance in single-turn versus multi-turn dialogues to evaluate whether multi-turn interactions enhance anomaly detection performance.

Experiment 3: impact of chain-of-thought step count on model performance

This experiment explores how varying the number of chain-of-thought steps affects the model's anomaly detection capabilities. By adjusting the number of reasoning steps, the experiment evaluates the model's performance at different depths of inference. The analysis focuses on the detection performance across various datasets to identify the optimal number of reasoning steps for the chain-of-thought strategy.

Experiment 4: performance comparison between small and larger models

To evaluate the performance gains of small models in anomaly detection tasks under the guided prompt framework, this experiment compares small models with larger language models (e.g., GPT-3.5-turbo, Llama-2-70b-chat, Mixtral-8x7B-Instruct). Using three benchmark datasets, the models are evaluated on four metrics to determine if small models, when guided by the prompt framework, can achieve or surpass the performance of larger models.

Experiment 5: impact of parameter combinations on model performance

This experiment investigates the influence of various parameter settings on the model's anomaly detection performance. Parameters such as do_sample, temperature, top_k, and top_p are adjusted to assess their effects on accuracy, precision, recall, and F1 score. The goal is to identify the optimal parameter configuration to enhance anomaly detection performance through tuning.

Analysis and discussion of experimental results

Impact of TAD-GP strategy on model performance

The results from Experiments 1, 2, and 3 (Table 1) demonstrate that the TAD-GP framework, incorporating strategies like chain-of-thought reasoning, multi-turn dialogue, and anomaly type introduction, significantly enhances anomaly detection performance in small models (Gemma-7b-it). By guiding the model to explore data

Dataset	Accuracy	Precision	Recall	F1 Score	Remarks
CICIDS2017	0.6600	**0.7222**	0.5200	0.6047	Experiment 1, 2[a]
CICIDS2017	**0.7600**	0.6970	**0.9200**	**0.7931**	Experiment 2, 3[b]
CICIDS2017	0.6800	0.6216	**0.9200**	0.7419	Experiment 3[c]
CICIDS2017	0.5000	0.000	0.000	0.000	Experiment 1[d]
KDD Cup 1999	0.7600	**1.0000**	0.5200	0.6842	Experiment 1, 2[a]
KDD Cup 1999	**0.9800**	**1.0000**	0.9600	**0.9796**	Experiment 2, 3[b]
KDD Cup 1999	0.7800	0.6944	**1.0000**	0.8197	Experiment 3[c]
KDD Cup 1999	0.5000	0.000	0.000	0.000	Experiment 1[d]
UNSW-NB15	0.4200	0.3750	0.2400	0.2927	Experiment 1, 2[a]
UNSW-NB15	0.6000	0.6471	0.4400	0.5238	Experiment 2, 3[b]
UNSW-NB15	**0.6400**	**0.6842**	**0.5200**	**0.5909**	Experiment 3[c]
UNSW-NB15	0.5000	0.000	0.000	0.000	Experiment 1[d]

Table 1. The Impact of Various TAD-GP Strategies on the Performance of the Gemma-7b-it Model. [a] Single turn dialogue + Example + Anomaly type + 5-step chain-of-thought. [b] Multi-turn dialogue + Example + Anomaly type + 5-step chain-of-thought. [c] Single turn dialogue. [d] Multi-turn dialogue + Example + Anomaly type + 3-step chain-of-thought. Significant values are in bold.

Model	Accuracy	Precision	Recall	F1 score
`Gemma-7b-it`	**0.7933**	**0.7937**	0.8000	**0.7879**
`GPT-3.5-turbo`	0.7533	0.7797	0.6667	0.7146
`Llama-2-70b-chat`	0.6267	0.6138	**0.9287**	0.7248
`Mixtral-8x7B-Instruct`	0.4267	0.3838	0.4980	0.4265

Table 2. Performance comparison between the Gemma-7b-it (TAD-GP) model and large-scale models. The table presents the average performance of each model across three datasets. For example, the Gemma-7b-it model's accuracy of 0.7933 represents the mean of its accuracy on the KDD Cup 1999 dataset (0.98), CICIDS2017 dataset (0.76), and UNSW-NB15 dataset (0.64). Significant values are in bold.

details step by step, the TAD-GP framework effectively addresses the challenges of identifying complex features in tabular data anomaly detection tasks.

The chain-of-thought strategy, in particular, improves anomaly detection by incrementally providing data features and enabling Chain-of-thought reasoning. For example, in the KDD Cup 1999 dataset, using 5-step chain-of-thought reasoning and multi-turn dialogue increased accuracy from 0.50 to 0.98, with an F1 score of 0.9796. This highlights the effectiveness of Chain-of-thought reasoning in capturing complex attack patterns. In contrast, single-turn dialogue resulted in suboptimal performance, suggesting that the absence of contextual information limits the model's ability to fully grasp data complexity.

However, the reasoning depth must align with the dataset's complexity. For instance, in the CICIDS2017 and KDD Cup 1999 datasets, 5-step reasoning outperformed 3-step reasoning. In the more complex UNSW-NB15 dataset, however, 3-step reasoning achieved an F1 score of 0.5909, slightly better than the 0.5238 achieved with 5-step reasoning. This suggests that excessive reasoning steps can introduce redundancy or noise, diminishing performance. Therefore, in practical applications, reasoning depth should be adjusted to balance accuracy and efficiency.

The multi-turn dialogue strategy was particularly effective in helping the model develop a comprehensive understanding of data features. This approach allows the model to progressively acquire critical information, improving anomaly identification. In the CICIDS2017 dataset, for instance, the F1 score increased from 0.6047 with single-turn dialogue to 0.7931 with multi-turn dialogue, emphasizing the importance of continuous context enhancement for understanding complex patterns. These findings suggest that TAD-GP's guided prompt strategies excel not only in detecting standard attack behaviors but also in identifying complex anomalies involving multiple dimensions and hidden patterns.

Comparison between small and larger models

Experiment 4 (Table 2) further highlights the advantages of the TAD-GP framework. In anomaly detection tasks, the TAD-GP framework enabled the small Gemma-7b-it model to outperform larger models, such as GPT-3.5-turbo, Llama-2-70b-chat, and Mixtral-8x7B-Instruct, across multiple performance metrics. Despite their larger parameter counts, these models did not match the performance of Gemma-7b-it without the optimization provided by the TAD-GP framework. This finding demonstrates that model effectiveness is not solely dependent on parameter size but is largely influenced by the application of guided prompt strategies.

do_sample	temperature	top_k	top_p	Accuracy	Precision	Recall	F1 score
FALSE	–	–	–	0.6400	0.6296	0.6800	0.6538
TRUE	0.9	50	0.95	0.6600	0.6818	0.6000	0.6383
TRUE	0.7	50	0.95	**0.7600**	0.6970	**0.9200**	**0.7931**
TRUE	0.5	50	0.95	0.7200	**0.7200**	0.7200	0.7200
TRUE	0.3	50	0.95	0.6200	0.6250	0.6000	0.6122
TRUE	0.7	120	0.95	0.7400	0.7308	0.7600	0.7451
TRUE	0.7	90	0.95	0.6800	**0.7692**	0.6667	0.7143
TRUE	0.7	50	0.95	**0.7600**	0.6970	**0.9200**	**0.7931**
TRUE	0.7	20	0.95	0.6600	0.6667	0.6400	0.6531
TRUE	0.7	50	0.95	**0.7600**	0.6970	**0.9200**	**0.7931**
TRUE	0.7	50	0.75	**0.7600**	**0.8095**	0.6800	0.7391
TRUE	0.7	50	0.5	0.6800	0.7368	0.5600	0.6364

Table 3. Impact of Parameter Combinations on Gemma-7b-it Model Performance. In Experiment 5, the performance of the Gemma-7b-it model, using the TAD-GP framework, was evaluated on the CICIDS2017 dataset under various parameter configurations for comparison. Significant values are in bold.

The Gemma-7b-it model achieved an F1 score of 0.7879, surpassing GPT-3.5-turbo's 0.7146 and Llama-2-70b-chat's 0.7248. This confirms that task-specific guidance via the TAD-GP framework can significantly enhance the performance of small models, sometimes exceeding that of larger models. This advantage is particularly evident in resource-constrained environments, where small models optimized with TAD-GP demonstrate superior efficiency. The TAD-GP framework not only offers competitive performance but also provides practical benefits in terms of reduced computational resource and energy consumption.

Interestingly, Mixtral-8x7B-Instruct performed significantly below expectations, with an F1 score of only 0.4265. This highlights that model size alone cannot compensate for the lack of a guided strategy when dealing with complex tabular data. In real-world applications, this underscores the importance of task-specific guidance, as parameter size alone does not guarantee superior performance.

Effect of parameter combinations on model performance

The results of Experiment 5 (Table 3) underscore the significant impact of parameter settings on model performance. Adjusting parameters such as temperature, top_k, and top_p led to substantial variations in accuracy, precision, recall, and F1 score. The optimal performance was achieved with a combination of temperature at 0.7, top_k at 50, and top_p at 0.95, resulting in an F1 score of 0.7931.

The experiment also reveals trade-offs between generation diversity and stability with different parameter settings. For instance, a higher top_k (e.g., 120) improved recall but introduced uncertainty, reducing precision. Therefore, parameter settings must be adjusted flexibly based on the data characteristics and task requirements to balance performance and stability. This highlights the significant optimization potential of generation model parameters across various tasks and scenarios.

Potential expanded applications of the TAD-GP framework

While this study focused primarily on anomaly detection in network security-related tabular data, the TAD-GP framework exhibits strong versatility and can be adapted to other data processing tasks. For instance, in financial fraud detection, where fraud tactics continuously evolve, traditional methods often struggle to detect new or mutated fraud patterns. The multi-turn dialogue and chain-of-thought strategies in the TAD-GP framework can effectively address this by enabling models to capture subtle changes in fraudulent behavior.

In medical diagnostics, the complexity and diversity of data often make abnormal patterns difficult to identify. The fine-grained anomaly type recognition offered by the TAD-GP framework can enhance models' ability to analyze diagnostic data, potentially improving early disease detection. Additionally, in supply chain management, where anomalies such as demand fluctuations and logistical bottlenecks occur, the TAD-GP framework can help detect these issues more accurately, thereby mitigating supply chain risks.

Challenges and future directions

Although the TAD-GP framework has demonstrated its advantages with structured tabular data, its potential with unstructured data remains underexplored. Given the success of large language models in multimodal tasks, such as natural language and image processing, future research could investigate extending the TAD-GP framework to handle more complex data types. Designing a multimodal prompt framework that integrates text, images, and speech, or developing cross-domain anomaly detection methods for multimodal tasks, could be promising avenues for exploration. This would not only broaden the TAD-GP framework's applications but also provide new approaches for tackling more challenging cross-modal anomaly detection tasks.

Additionally, while the TAD-GP framework has shown strong generalization capabilities through few-shot learning, its performance may be affected by domain-specific data distribution differences. Future research could explore fine-tuning pre-trained models with small amounts of domain-specific data to improve performance in particular fields. By implementing domain-adaptive fine-tuning strategies, the TAD-GP framework could

better accommodate the characteristics of new domains, reducing false positives and negatives and enhancing its practical utility in real-world applications.

Conclusion

The TAD-GP (Tabular Anomaly Detection via Guided Prompts) framework proposed in this paper offers an innovative and practical approach to improving the performance of small-scale language models in detecting anomalies in tabular network data. By leveraging strategies such as chain-of-thought reasoning, multi-turn dialogue, example guidance, anomaly type recognition, and information reinforcement, TAD-GP significantly outperforms larger models like GPT-3.5-turbo, Llama-2-70b-chat, and Mixtral-8x7B-Instruct on datasets such as KDD Cup 1999, CICIDS2017, and UNSW-NB15.

This highlights the substantial potential of guided prompt strategies for optimizing model performance, especially in resource-constrained environments that require high security and privacy. The TAD-GP framework allows small models to achieve effective anomaly detection while offering advantages in energy efficiency, flexible deployment, and privacy protection compared to larger models. By employing natural language-based guided prompts, the framework eliminates the need for complex feature engineering, demonstrating strong adaptability and robustness, making it highly practical in real-world applications.

Beyond enhancing small models' anomaly detection capabilities, the TAD-GP framework opens new possibilities for efficient and flexible anomaly detection in tabular data. Future research could further validate the framework's generalizability and explore its potential in fields such as finance, healthcare, and supply chain management. This would advance the development of efficient, secure anomaly detection technologies. Widespread adoption of the TAD-GP framework promises cost-effective, high-performance solutions for anomaly detection across diverse domains.

Data availibility

The datasets employed in the preparation of this paper are publicly accessible via the following URLs: KDD Cup 1999 dataset (20,000 records): https://www.kaggle.com/datasets/lengxingxin/kdd99-20000-json CICIDS2017 dataset (20,000 records): https://www.kaggle.com/datasets/lengxingxin/cicids2017-20000-json UNSW-NB15 dataset (20,000 records): https://www.kaggle.com/datasets/lengxingxin/unsw-nb15-20000-json

Code availability

The code implemented in this study is available for public access at the following URLs: Code for the small model: https://www.kaggle.com/code/lengxingxin/code-for-small-model Code for the large model: https://www.kaggle.com/code/lengxingxin/code-for-large-model.

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Author contributions

X.Z. conceived and designed the study. X.Z. and X.L. conducted the experiments. X.L. and L.W. analyzed the data. N.W. and Y.L. contributed to data interpretation. All authors reviewed and approved the final manuscript.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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